SSLC EXAMINATION , MARCH - 2026

MATHEMATICS – ANSWER KEY

S 3031

Qn no	Key	Score
	SECTION A	
1	(b) 7 , 12 , 17 , . . .	1
2	(d) 20 square centimetres	1
3	(b) (0,2)	1
4	(a) 42	1
5	(d) $4n + 2$	1
6	(b) 4 centimetres	1
7	(c) Both **statements** are true , **statement** 2 is the reason of **statement** 1.	1
8	(c) (ii) and (iii) are true	1
	SECTION B	
9	Central angle of the shaded region $= 360 - 120 = 240^0$ Probability that the dot is in the shaded region $= \frac{240}{360} = \frac{2}{3}$	1 1
10A	Total number of three digit numbers $= 900$ (i) Probability that all the three digits are the same $= \frac{9}{900} = \frac{1}{100}$ Three digit multiples of 5 less than 200 $= 100 ,105 ,110 ,115 ,120,125, 130,135,140,145, 150,155,160,165,170,175,180,185,190,195$ (ii) Probability that the three digit number is multiple of 5 less than 200 $= \frac{20}{900} = \frac{1}{45}$	1 1 1
10B	Total number of pairs $= 4 \times 3 = 12$ (i) Probability that the product of the numbers being a prime $= \frac{1}{12}$ (ii) Probability that the product of the numbers is a multiple of 7 $= \frac{3}{12} = \frac{1}{4}$	1 1 1
11	(i) $x_{15} = 140 + 40 = 180$ OR $x_{15} = 100 + (2 \times 40) = 180$ (ii) $x_3 = 100 - 40 = 60$ OR $x_3 = 140 - (2 \times 40) = 60$	1 1
12A	(i) $x_2 = \frac{30}{3} = 10$ (ii) $x_4 = \frac{140}{7} = 20$ (iii) Common difference $= \frac{10}{2} = 5$ First three terms $= 5 , 10 , 15$	1 1 1 1
12B	(i) $x_4 + x_{22} = x_1 + x_{25} = 70$ (ii) $x_{13} = \frac{70}{2} = 35$ $[x_{12} + x_{14} = 70]$ (iii) Sum of the first 25 terms $= 25 \times x_{13} = 25 \times 35 = 875$	1 1 2

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Score	Number of students
Below 40	6
Below 50	11
Below 60	19
Below 70	29
Below 80	38
Below 90	45

$N = 45$

(i) Median score = Score of the 23rd student $= x_{23}$ — 1

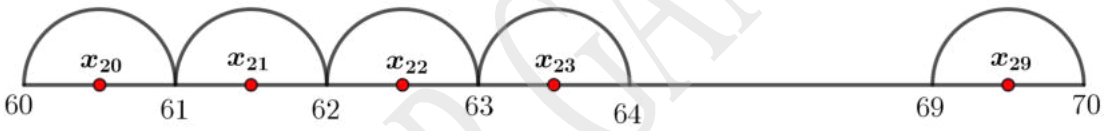
Median class $= 60 - 70$

There are 10 students in the median class — 1

Divide the 10 scores between 60 and 70 into 10 equal parts .

Length of one sub division $= \frac{10}{10} = 1 = d$ — 1

Assume that each such subdivision contains one student whose score is the mid value of that subdivision . .



(ii) Score of the 20th student $= x_{20} = \frac{60 + 61}{2} = 60.5$ — 1

The scores in the median class are in arithmetic sequence

(iii) Median score $= x_{23} = 60.5 + (3 \times 1) = 63.5$ — 1

SECTION C

14

(i) Coordinates of the centre of the circle $= \left(\frac{4 + 8}{2}, \frac{3 + 5}{2}\right) = (6, 4)$ — 1

(ii) Coordinates of the another end of the diameter $= ([2 \times 6] - 5, [2 \times 4] - 2)$ — 1

$= (7, 6)$ — 1

15

(i) Coordinates of C $= (4, 0)$ — 1

(ii) Height from A to BC $= 2\sqrt{3}$ — 1

(iii) Coordinates of A $= (2, 2\sqrt{3})$ — 1

16A

(i) Coordinates of the centre of the circle $= \left(\frac{3 + 5}{2}, \frac{5 + 9}{2}\right) = (4, 7)$ — 1

(ii) Radius of the circle $= \sqrt{(4-3)^2 + (7-5)^2} = \sqrt{5}$ — 1

(iii) $(x-4)^2 + (y-7)^2 = 5$ — 2

16B

(i) Slope of a line $= \frac{7-5}{9-3} = \frac{2}{6} = \frac{1}{3}$ — 1

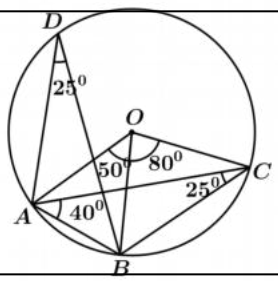
(ii) If we take (x,y) as a point on this line , $\frac{y-5}{x-3} = \frac{1}{3}$ — 2

$x - 3y + 12 = 0$

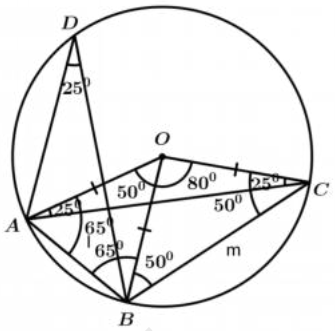
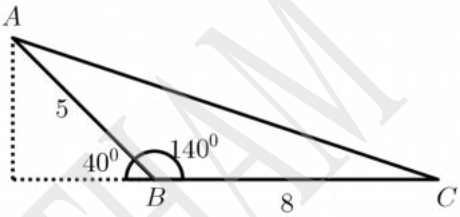
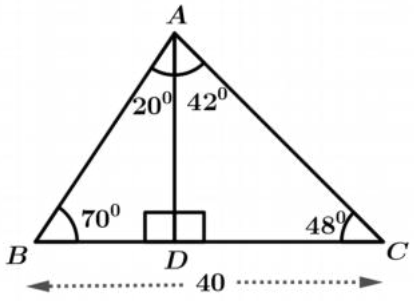
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	(iii) $(6\ ,\ 6)$ [Any point got from the given points such that as the change in x coordinates is 3 , then the change in y is 1]	1
	SECTION D	
17A	(i) If we take the length of the shorter side as x *m*, then the length of the longer side is $x+30$ *m*.	1
	$x(x+30) = 351 \quad ==> \quad x^2 + 30x = 351$	
	$x^2 + 30x + 15^2 = 351 + 15^2$	1
	$(x+15)^2 = 576$	
	$x + 15 = \sqrt{576} = 24 \quad ===> \quad x = 9$	1
	Length of the shorter side = 9 m.	
	OR	
	(ii) $x^2 + 30x = 351 \quad ==> \quad x^2 + 30x - 351 = 0$	
	$x = \dfrac{-30 \pm \sqrt{30^2 - 4\times 1\times(-351)}}{2\times 1} = \dfrac{-30 \pm \sqrt{2304}}{2}$	
	$x = \dfrac{-30+48}{2}$ OR $x = \dfrac{-30-48}{2}$	
	$x = 9$ OR $x = -39$	
	Length of the shorter side = 9 m.	
17B	If we take the numbers as x and y , $x + y = 4$, $xy = 2$	
	$(x-y)^2 = 4^2 - (4\times 2) = 8$	1
	$x - y = \sqrt{8}$	1
	$x = \dfrac{4+\sqrt{8}}{2}$, $y = \dfrac{4-\sqrt{8}}{2}$	1
	$\left[\ x = \dfrac{4+2\sqrt{2}}{2} = 2+\sqrt{2} \quad , \quad y = \dfrac{4-2\sqrt{2}}{2} = 2-\sqrt{2}\ \right]$	
	OR	
	If we take the numbers as $2+x$ and $2-x$,	
	$(2+x)(2-x) = 2$	
	$4 - x^2 = 2 \quad ==> \quad x^2 = 2 \quad ==> \quad x = \sqrt{2}$	
	Numbers $= 2+\sqrt{2}$, $2-\sqrt{2}$	
	OR	
	If we take first number as then the other number is $4 - x$	
	$x(4-x) = 2 \quad ==> \quad 4x - x^2 = 2 \quad ==> \quad x^2 - 4x = -2$	
	$x^2 - 4x + 2^2 = -2 + 2^2$	
	$(x-2)^2 = 2$	
	$x - 2 = \sqrt{2} \quad ==> \quad x = 2+\sqrt{2}$ OR $x = 2-\sqrt{2}$	
	Numbers $= 2+\sqrt{2}$, $2-\sqrt{2}$	
18	(i) 1 , 5 , 9 , . . .	1
	(ii) Yes , it is an arithmetic sequence.	1
	Since this sequence starting with 1 and proceeding by adding 4 again and again.	1
	(iii) $4n - 1$	1
	OR	

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	(ii) Yes , it is an arithmetic sequence. If we take any two consecutive terms of this sequence, the first would be 1 added to 4 multiplied by a number and the second, 1 added to 4 multiplied by the next number. So, the first subtracted from the second would give 4.	
19A	(i) Sum of the first 7 terms $= 7^2 + (6 \times 7) = 91$ (ii) $n^2 + 6n = 315$ $n^2 + 6n + 3^2 = 315 + 3^2 \implies (n+3)^2 = 324$ $n + 3 = \sqrt{324} = 18 \implies n = 15$ Number of terms $= 15$ OR (ii) $n^2 + 6n = 315 \implies n^2 + 6n - 315 = 0$ $n = \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times (-315)}}{2 \times 1} = \frac{-6 \pm \sqrt{1296}}{2}$ $n = \frac{-6 + 36}{2} = 15$ OR $n = \frac{-6 - 36}{2} = -21$ Number of terms $= 15$	1 1 1 1
19B	(i) $6n + 5 - 6 = 6n - 1$ (ii) $x_8 = (6 \times 8) - 1 = 47$ Sum of the first 15 terms $= 15 \times 47 = 705$ (iii) Sum of the first n terms $= 6 \times \frac{n(n+1)}{2} - n = 3n^2 + 2n$ OR Sum of the first 15 terms $= 6 \times \frac{15 \times 16}{2} - 15 = 705$	1 1 1 1
20	(i) If we take $x^2 - 12x + 32 = (x+a)(x+b)$, then $x^2 - 12x + 32 = x^2 + (a+b)x + ab$ $a + b = -12$, $ab = 32$ $(a-b)^2 = (-12)^2 - (4 \times 32) = 16$ $a - b = \sqrt{16} = 4$ $a = \frac{-12 + 4}{2} = -4$, $b = \frac{-12 - 4}{2} = -8$ $\therefore\ x^2 - 12x + 32 = (x-4)(x-8)$ (ii) 4 , 8 OR (i) -4 and -8 can be found directly and can be write $x^2 - 12x + 32 = (x-4)(x-8)$ directly. [Can be done using any correct method]	1 1 1 1 1
	SECTION E	
21	(i) $\angle ADB = \frac{1}{2} \times 50 = 25^0$ (Half the central angle of the small arc AB) (ii) $\angle BAC = \frac{1}{2} \times 80 = 40^0$ (Half the central angle of the small arc BC) 	1 1 1

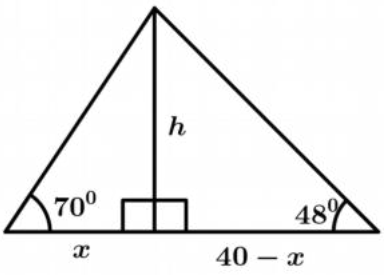
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	$\angle ACB = \frac{1}{2} \times 50 = 25^0$ (Half the central angle of the small arc AB) $\angle ABC = 180 - (40 + 25) = 115^0$ OR (ii) To identify that the triangle AOB and BOC are isosceles $\angle ABC = 65 + 50 = 115^0$ $\angle BAC = 65 - 25 = 40^0 \implies \angle ACB = 25^0$ 	1
22	Perpendicular distance from A to BC $= 5 \times \sin 40^0$ $= 5 \times 0.6428\ cm$ Area of the triangle ABC $= \frac{1}{2} \times 8 \times 5 \times 0.6428$ $= 12.856\ sq.cm$ 	2 2
23A	Height $= \sqrt{10^2 - 6^2} = 8\ cm$ Volume of the cylinder $= \pi \times 8^2 \times 15 = 960\pi\ cubic.cm$ Volume of the cone $= \frac{1}{3} \times \pi \times 6^2 \times 8 = 96\pi\ cubic.cm$ Number of cones $= \frac{960\pi}{96\pi} = 10$	1 1 1 2
23B	Slant height $= \sqrt{\left(\frac{18}{2}\right)^2 + 12^2} = 15\ cm$ Surface area of a square pyramid $= 18^2 + (2 \times 18 \times 15) = 864\ sq.cm$ $= \frac{864}{10000}\ sq.m$ Total cost $= 500 \times \frac{864}{10000} \times 120 = 5184\ Rs$	1 1 1 2
24A	(i)  (ii) If we take the height of the tower as h m, In the smaller right triangle , length of the shorter side $= h \times \tan 20^0\ m$ In the larger right triangle , length of the shorter side $= h \times \tan 42^0\ m$ $h \times \tan 20^0 + h \times \tan 42^0 = 40$ $h = \frac{40}{\tan 20^0 + \tan 42^0} = \frac{40}{0.364 + 0.9004} = \frac{40}{1.2644}\ m$	1 1 1 1 1

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OR

(i)



(ii) If we take the height of the tower as h m ,

In the smaller right triangle , $h = x \times \tan 70^0$ m

In the larger right triangle , $h = (40 - x)\tan 48^0$ m

$x\tan 70^0 = (40 - x)\tan 48^0$

$x\tan 70^0 + x\tan 48^0 = 40\tan 48^0 \implies x(\tan 70^0 + \tan 48^0) = 40\tan 48^0$

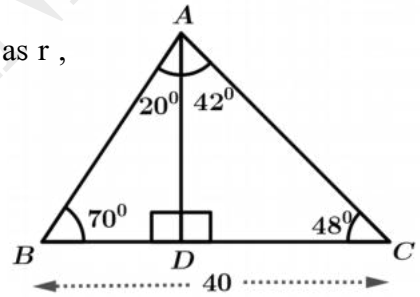
$$x = \frac{40\tan 48^0}{\tan 70^0 + \tan 48^0} \ m$$

$$\therefore \quad h = x \times \tan 70^0 = \frac{40\tan 48^0}{\tan 70^0 + \tan 48^0} \times \tan 70^0$$

$$= \frac{40 \times 1.1106}{2.7475 + 1.1106} \times 2.7475 = \frac{40 \times 1.1106 \times 2.7475}{3.8521} \ m \ .$$

OR

If we take the radius of the circumcircle of the triangle ABC as r ,



then $2r = \dfrac{40}{\sin 62^0}$ m

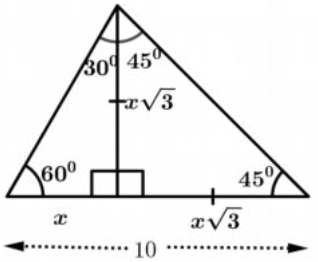
In triangle ABC , $AB = 2r\sin 48^0 = \dfrac{40}{\sin 62^0} \times \sin 48^0$ m

In triangle ABD , $AD = AB \times \sin 70^0$

$$= \frac{40}{\sin 62^0} \times \sin 48^0 \times \sin 70^0$$

$$= \frac{40}{0.8829} \times 0.7431 \times 0.9397 \ m \ .$$

24B If we take the shortest side of the smaller triangle as x cm , then



$x + x\sqrt{3} = 10 \implies x(1 + \sqrt{3}) = 10$ — 1

$$x = \frac{10}{1 + \sqrt{3}}$$ — 1

Height towards the side of length 10 cm $= x\sqrt{3}$

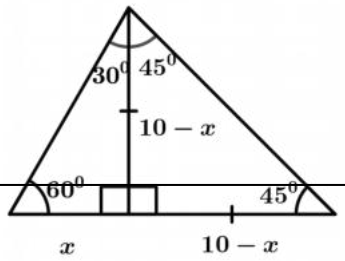
$$= \frac{10\sqrt{3}}{1 + \sqrt{3}} \ cm$$ — 1

Area of the triangle $= \dfrac{1}{2} \times 10 \times \dfrac{10\sqrt{3}}{1 + \sqrt{3}}$

$= 25\sqrt{3}(\sqrt{3} - 1) = 75 - 25\sqrt{3}$ $sq.cm$ — 2

OR

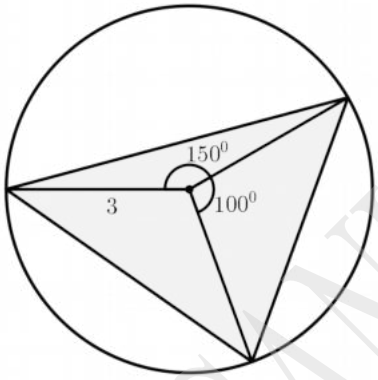
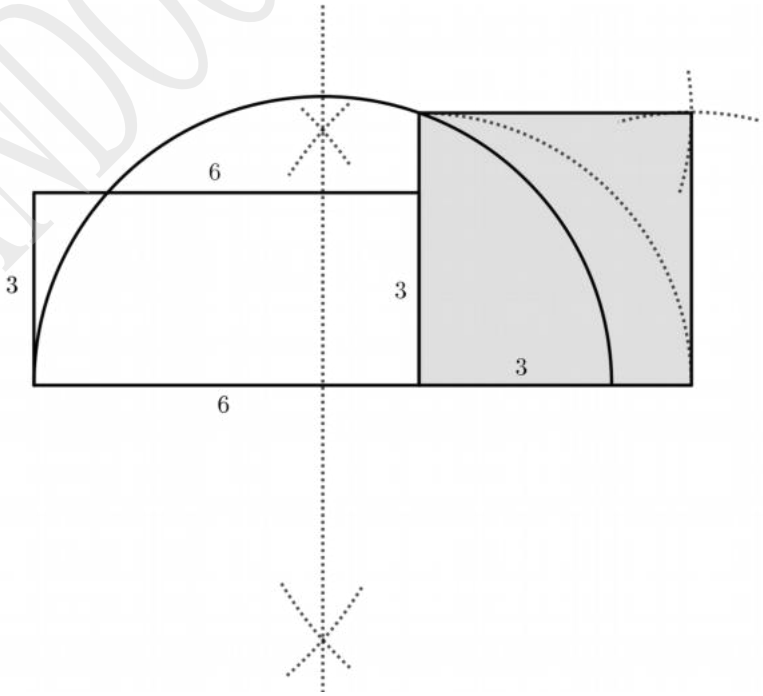
If we take the shortest side of the smaller triangle as x cm , then



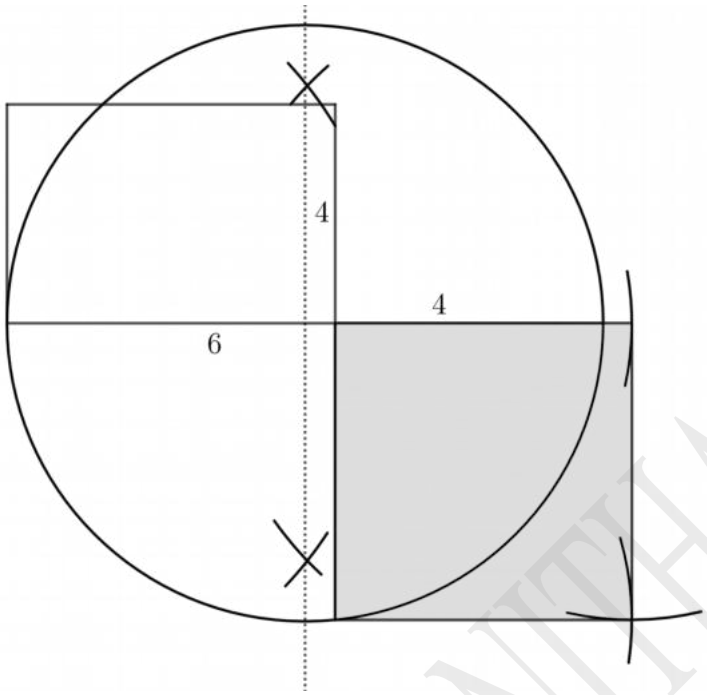
$10 - x = x\sqrt{3}$

$x + x\sqrt{3} = 10 \implies x(1 + \sqrt{3}) = 10$

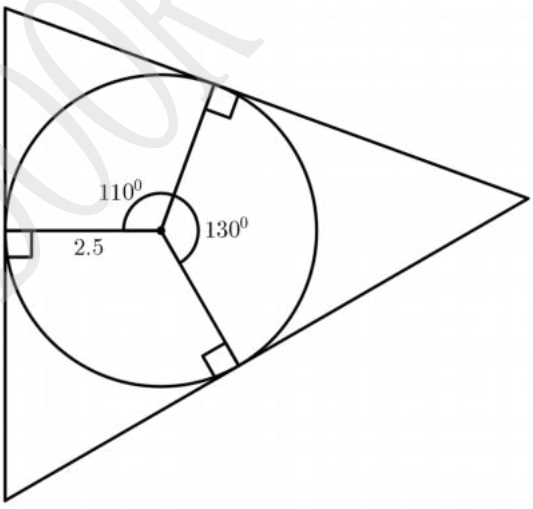
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	$x = \frac{10}{1 + \sqrt{3}}$ Height towards the side of length 10 cm $= x\sqrt{3}$ $= \frac{10\sqrt{3}}{1 + \sqrt{3}}\ cm$ Area of the triangle $= \frac{1}{2} \times 10 \times \frac{10\sqrt{3}}{1 + \sqrt{3}}$ $= 25\sqrt{3}(\sqrt{3} - 1) = 75 - 25\sqrt{3}\ sq.cm$	
25	For drawing a circle of radius 3cm For taking double the angles of the triangle at the centre of the circle For completing the triangle  	1 2 1
26	For drawing the rectangle. For extending the longer side by the shorter side . For drawing a circle with line as diameter. For completing the square.  	1 1 1 1

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27	For drawing a circle of radius 2.5 cm	1
	For subtracting each angle of the triangle from 180^0 and taking those angles at the centre of the circle	2
	For drawing tangents and completing the triangle	2



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